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SYSTEM DESIGN DOCUMENT DOST STARBOOKS: WHIZ CHALLENGE

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APRIL 2026





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INTRODUCTION

This System Design Document has been created to outline the proposed system design for the STARBOOKS Whiz Challenge Version 1.0; a gamified educational platform developed for the Department of Science and Technology – Science and Technology Information Institute (DOST-STII). The STARBOOKS Whiz Challenge is a ground-up redevelopment of the existing DOST-STII Whiz Challenge application, which was limited to a single solo quiz format with minimal engagement features and no administrative management tools.

This new version addresses the identified limitations of the legacy application by introducing four distinct game modes (Whiz Challenge, Whiz Memory Match, Whiz Puzzle, and Whiz Battle), a comprehensive rewards and leaderboard system, and a fully featured web-based administrative dashboard. The system is designed to operate in an offline, Local Area Network (LAN) environment to ensure accessibility in DOST event venues without reliable internet connectivity. This document and the technical specifications described herein have been developed in alignment with the project's Software Requirements Specification and the development standards agreed upon with DOST-STII.

PURPOSE

The purpose of this System Design Document is to provide a clear and comprehensive description of how the STARBOOKS Whiz Challenge system will be constructed, structured, and deployed. It translates the functional and non-functional requirements defined in the Software Requirements Specification (SRS) into concrete architectural, hardware, software, database, and security design decisions.

This document serves as the authoritative technical reference for the development team and a communication tool for DOST-STII stakeholders, faculty advisers, and future maintainers of the system. It ensures that the system design aligns with DOST-STII's mission to make quality STEM education accessible, engaging, and measurable for students from elementary through senior high school. It also establishes the design framework necessary to support future enhancements and content updates by DOST-STII technical staff after project handover.

SYSTEM OVERVIEW

The STARBOOKS Whiz Challenge Version 1.0 is a standalone, LAN-based gamified learning platform designed to complement DOST-STII's existing STARBOOKS initiative. The system targets students attending DOST events, science fairs, and public exhibits, offering interactive STEM learning through multiple game modes that accommodate different learning styles and promote sustained engagement.



The system is architected as a three-component client-server application deployed within an isolated local area network. The three primary components are a Flutter/Dart web-based game client accessed by players through standard browsers, a web-based administrative dashboard for DOST-STII staff, and a centralized backend server running the Laravel PHP framework with a MongoDB database. All components communicate exclusively within the LAN environment, with no internet connectivity required during deployment at events.

At a high level, the system supports two primary user roles. Players can register accounts, select from four game modes, earn badges and stars, view leaderboards, and submit reward claims. Administrators can manage player accounts, configure game settings and question content, monitor reward redemptions, and access a comprehensive analytics dashboard covering player demographics, game mode preferences, and reward distributions.

The system is designed to support a minimum of 10 concurrent active player sessions while maintaining acceptable response times under standard LAN conditions. It is intended to be self-contained and portable, deployable at any DOST venue by transferring the application and starting the Docker-based server services with no internet downloads or complex installations required on-site.

DESIGN CONSTRAINTS

The STARBOOKS Whiz Challenge development team identified several constraints that directly influence and limit the system design. These constraints arise from the operational environment, organizational requirements, technology selections, and regulatory obligations, and must be carefully factored into every aspect of the system architecture.

Hardware Constraints. The system must operate on standard Windows-based personal computers commonly available at DOST event venues. Minimum hardware specifications include an Intel Core i3 or equivalent processor, 4GB RAM, and at least 50GB of available storage. The system design cannot assume higher-end hardware, which places limits on computational complexity, in-memory data processing, and the number of concurrent users the server can reliably support. The design targets a maximum of 15 concurrent users (players and administrators combined) under these constraints.

Network Constraints. The system must operate exclusively within an offline LAN environment. All client-server communication occurs through a local router using standard Ethernet connectivity (100 Mbps or higher). The design cannot rely on any cloud services, external APIs, or internet-based resources during event operation. This constraint drives the requirement for self-contained deployment packages and local data storage through MongoDB.



Technology Constraints. The frontend must be developed using the Flutter/Dart framework compiled for web deployment. The backend must use the Laravel PHP framework running on the Laragon development stack (or Docker for production deployment). The database must be MongoDB. These technology selections are fixed requirements that constrain the design team from using alternative frameworks or databases, and all design decisions must remain compatible with these specified technologies.

Regulatory Constraints. The system must comply with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173) regarding the collection, storage, and processing of personal information from players, the majority of whom are minors. This restricts what data can be collected to the minimum necessary (username, age, gender, school, region, province, city), prohibits collection of sensitive personal information such as full names or contact numbers, and requires that data remain stored locally within DOST-controlled infrastructure.

Organizational Constraints. The complete system, including source code, documentation, and deployment instructions, must be delivered to DOST-STII upon project completion. The design must therefore prioritize maintainability, with well-documented modular code that DOST-STII technical staff can manage without involvement from the original development team. The administrative interface must also be usable by non-technical DOST staff for routine content management tasks such as adding questions and reviewing analytics.

Content Constraints. All educational content including quiz questions, correct answers, difficulty classifications, subject categories, and puzzle images must be provided by and validated by DOST-STII content specialists. The system design accommodates this dependency by separating content management from the application logic through the administrative dashboard, enabling DOST staff to independently manage and update content post-deployment.



ROLES AND RESPONSIBILITIES

The following table defines the MMS System Design roles and responsibilities. This matrix also serves as the list of points of contact for issues and concerns relating to the MMS System Design.

Name	Role	Responsibility
Kelly Dumbrique	Project Manager / Product Owner	Overall project oversight; client communication with DOST-STII; administrative dashboard frontend development
Arcielle Marie Gercan	Backend Developer	Server-side API development using Laravel; database integration; WebSocket server for Whiz Battle
Shandrae Lois Quianzon	Frontend Developer	Player-side game client development using Flutter/Dart; Whiz Challenge and Homepage implementation
Janice Maxene Salipande	Scrum Master / QA Lead	Whiz Puzzle and Whiz Memory Match implementation; system diagrams; OpenProject tracking
Mr. Jose Eugenio Quesada	Faculty Adviser	Project guidance; sprint reviews; design feedback
Mavie Narquita Mc Gyver Basaya Jonathan Abalon Ella Pestano	DOST-STII Stakeholders	Requirements validation; content provision; sprint reviews; final acceptance

PROJECT REFERENCES

The STARBOOKS Whiz Challenge system design is informed and governed by the following reference documents. These references serve as the basis for requirements, design decisions, standards compliance, and operational guidance. Documents marked as living references may be periodically updated and should be consulted individually for the most current information.

- Software Requirements Specification (SRS) – STARBOOKS Whiz Challenge Version 1.0, February 24, 2026
- STARBOOKS Whiz Challenge User Manual (to be delivered upon project completion)
- Philippine Data Privacy Act of 2012, Republic Act No. 10173 – National Privacy Commission
- Department of Education K-12 STEM Curriculum Standards – for educational content alignment



- DOST-STII STARBOOKS Project Documentation – for branding guidelines and educational content standards
- Flutter/Dart Framework Documentation – for frontend development standards
- Laravel PHP Framework Documentation – for backend API development standards
- MongoDB Documentation – for database design and query standards
- Sprint Review and Retrospective Meeting Minutes (Sprints 3–5), November 2025 – February 2026

SYSTEM ARCHITECTURE

The STARBOOKS Whiz Challenge system is built on a client-server architecture designed for offline LAN-based deployment. All components operate within an isolated local area network, with no internet connectivity required during events.

Hardware:

The hardware architecture consists of two types of nodes: a centralized server for PC and multiple client PCs — all interconnected through a LAN router.

The **Server PC** hosts all backend services and acts as the central node for the entire system. It requires the following:

- An Intel Core i3 or equivalent processor, a minimum of 4GB RAM, at least 50GB of available storage, and a Network Interface Card (NIC) supporting Ethernet connectivity.
- The server runs the Docker Engine as its execution environment, which orchestrates three containerized services defined through a docker-compose.yml configuration file: the Laravel Backend API container, the WebSocket Server container, and the MongoDB Database container.

Client PCs serve as player and administrator terminals. Each requires the same minimum hardware specifications as the server:

- Intel Core i3, 4GB RAM, 50GB storage
- A display monitor, a standard USB keyboard and mouse, and a network interface card for LAN connectivity.
- Players and administrators must access the system exclusively through a web browser (Chrome or Edge).

The **LAN Router** provides network connectivity between the server and all client machines using standard Ethernet (100 Mbps or higher), with sufficient ports to accommodate all connected PCs.



Software:

The software architecture is organized into three primary components:

- The **Flutter/Dart Web Application (Game Client)** runs within the browser on each client PC. It serves as the presentation layer for both the player-facing game interface and the administrative dashboard. The web application communicates with the Laravel backend through RESTful HTTP API calls for standard operations, and via WebSocket connections for real-time Whiz Battle functionality. Key application artifacts include `main.dart`, `login.dart`, `home.dart`, `game.dart`, and `admin_dashboard.dart`, along with associated asset folders for images, sounds, and fonts.
- The **Laravel Backend API** is containerized and runs on the server PC and handles all business logic including user authentication and session management, game state processing, reward and badge management, leaderboard ranking, analytics data aggregation, and administrative content management. It exposes RESTful API endpoints consumed by the Flutter frontend and manages all interactions with the MongoDB database.
- The **WebSocket Server** is a separate containerized component running alongside Laravel on the server. It manages real-time bidirectional communication specifically for the Whiz Battle game mode, enabling two players to create and join battle rooms using generated game codes, synchronize question delivery, and exchange real-time score updates.

DATABASE DESIGN

The STARBOOKS Whiz Challenge system uses **MongoDB** as its primary database management system. MongoDB's document-oriented, flexible schema design is well-suited to the system's varied data structures, which range from player profiles and game session records to question banks and analytics data.

All data is stored locally on the server PC within the MongoDB container's persistent data/db/volume, ensuring that player records and game data are retained across system restarts and between event days. The database communicates with the Laravel backend through native MongoDB PHP drivers, which handle all CRUD operations.

The database is organized around the following primary collections, as defined in the Entity Relationship Diagram included in the SRS:

PLAYER_INFO stores all registered player profile data including username, hashed password, age, sex, school, student category, region, province, city, selected avatar, and account timestamps. This collection is the central reference point for player identity across all other collections.



PLAYER_STATS tracks individual player gameplay statistics including total session counts, game mode preferences, and session durations, feeding into the analytics dashboard.

PLAYER_BADGES records all badges earned by each player, organized by game mode (Whiz Challenge and Whiz Battle), difficulty level (Easy, Average, Difficult), and category. This collection drives both the reward redemption eligibility check and the leaderboard ranking system.

PLAYER_STARS stores star-based performance records from Whiz Memory Match and Whiz Puzzle sessions, including performance metrics used for leaderboard display.

PLAYER_REWARDS logs all reward claim transactions, capturing player ID, difficulty level claimed, claim timestamp, claim status (pending or granted), and the administrator who processed the grant.

QUIZ_QUESTIONS contains the full question bank for Whiz Challenge and Whiz Battle, with each document storing question text, four answer options, the correct answer designation, subject category (Science or Math), and difficulty level.

QUIZ_DIFFICULTY_SETTINGS stores configurable game parameters per difficulty level, including the number of questions per session and the time limit per question, as managed through the administrative dashboard.

BATTLE stores active and completed Whiz Battle session records, including battle room codes, participating player IDs, question sets, individual scores, and outcome data.

FASTEST_TIME_MEMORY_MATCH and **FASTEST_TIME_PUZZLE** record best completion times for each player in their respective game modes, used for personal and overall leaderboard display.

ANALYTICS serves as an aggregated data store supporting the analytics dashboard visualizations, including total player counts, game mode usage frequency, demographic distributions, and reward distribution patterns.

ADMIN_INFO stores administrator account records including username and hashed password, with role-based access flags distinguishing admin accounts from player accounts. All password fields across player and admin collections are stored using strong one-way cryptographic hashing with unique salts, in compliance with the system's security requirements. The database schema is fully documented in the SRS's Entity Relationship Diagram and the



associated data dictionary, which provide field-level descriptions, data types, and collection relationships.

HARDWARE AND SOFTWARE DETAILED DESIGN

Hardware:

The STARBOOKS Whiz Challenge system leverages standard Windows-based PC hardware available at DOST event venues. No custom or proprietary hardware is required. The server PC and client PCs share the same minimum hardware profile (Intel Core i3, 4GB RAM, 50GB storage) to simplify procurement and compatibility management for DOST-STII.

The server PC is distinguished by its role, not its hardware configuration — it is the machine on which Docker Engine is installed and from which the three backend containers are launched. Client PCs require only a compatible web browser (Chrome or Edge with HTML5 and JavaScript support) to access the system. The LAN router connects all devices using standard Ethernet cabling and requires no specialized configuration beyond standard IP address assignment within the local network.

Software:

The STARBOOKS Whiz Challenge software is custom-developed by Group NEXUS to meet the specific educational, operational, and engagement requirements of DOST-STII. No commercial off-the-shelf (COTS) product was identified that could adequately address the combination of offline deployment, multi-game functionality, rewards management, and STEM content administration required by the client.

The software is organized into the following modules:

Player Account Management Module handles all player-facing authentication and profile operations. It is coded to support multi-step registration (privacy notice acknowledgment followed by profile information entry), credential-based login with session management, profile editing, password change with current password verification, and session logout. Input validation is enforced at both the frontend (Flutter) and backend (Laravel) layers for all account operations.

Game Engine Module encompasses the four game mode implementations. The Whiz Challenge sub-module manages question retrieval from the database based on selected category and difficulty, countdown timer operation, answer validation with immediate visual feedback, score calculation, and badge award logic upon perfect score achievement. The Whiz Memory Match sub-module manages card grid generation with randomized image pairs, card flip logic, match detection, elapsed time tracking, move counting, and star-based scoring. The Whiz Puzzle sub-module manages image selection and jigsaw piece generation by difficulty level, drag-and-drop piece movement, proximity-based snap detection, and completion validation. The Whiz Battle



sub-module manages unique game code generation, player matchmaking through code entry, synchronized question delivery to both players via WebSocket, real-time score display, and badge award for the winning player.

Reward Redemption Module manages the full badge-to-reward pipeline. It enforces the eligibility rule that a player must accumulate exactly three badges within a single difficulty level before a claim request is permitted. Player-side claim submission, administrator-side claim verification and grant processing, badge count reset upon successful grant, and permanent transaction logging are all handled within this module.

Administrative Content Management Module provides DOST-STII staff with tools to manage the question bank (add, edit, delete questions with full field validation), configure game parameters per difficulty level (number of questions and time limit per question), manage player and administrator accounts, and process reward claims.

Analytics and Leaderboard Module aggregates player data across all collections to generate the visualizations displayed on the administrative dashboard, including gender distribution, age distribution, regional distribution, game mode popularity by demographic, and reward distribution patterns. Leaderboard rankings are computed in real time based on total rewards claimed, with the most recent claim timestamp used as a tiebreaker. Both analytics data and leaderboard data support export functionality for DOST-STII reporting purposes.

SYSTEM SECURITY AND INTEGRITY CONTROLS

The STARBOOKS Whiz Challenge system incorporates a multi-layered approach to security and data integrity, designed to protect player data, particularly that of minors, and ensure consistent, reliable system operation during DOST events.

User Authentication and Role-Based Access Control. All system users, both players and administrators, must authenticate with valid username and password credentials before accessing any system functionality. The system maintains a strict separation between player accounts and administrator accounts through separate authentication pathways and role-based access control. Players are restricted to game functionality and their own profile management. Administrators have exclusive access to the management dashboard, analytics, content configuration, and reward processing. No player account, regardless of credentials, can access administrative functions.

Password Security. All passwords are stored using strong one-way cryptographic hashing algorithms with unique salts per user account. Plaintext and reversible encrypted password storage are explicitly prohibited in the system design. Password creation is enforced to require



a minimum of eight characters, including a combination of uppercase and lowercase letters, numbers, and special characters.

Data Privacy Compliance. The system is designed in full compliance with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173). A mandatory Privacy Notice is displayed and must be acknowledged during player registration before any personal information is entered. Data collection is limited to the minimum fields necessary for system functionality: username, age, gender, school, region, province, and city. No sensitive personal information such as full names, contact numbers, or email addresses is collected. All collected data is stored exclusively within DOST-controlled local infrastructure and is never transmitted to external parties.

Network Isolation. The system operates on an isolated LAN with no internet connectivity during events. All client-server communication occurs within the closed local network. The server exposes only the necessary HTTP/HTTPS and WebSocket ports for client communication. All other network ports are closed or filtered to minimize the attack surface. MAC address filtering or equivalent access control mechanisms on the LAN router to prevent unauthorized devices from joining the network.

Data Integrity and Audit Controls. The system enforces input validation at both the frontend and backend layers for all data entry operations, preventing malformed or incomplete data from being saved to the database. Confirmation dialogs are required before any destructive operations such as account deletion or question removal. All reward claim transactions are permanently logged with timestamps, player identity, difficulty level, and administrator identity, providing a complete audit trail of the reward system.

Operational Stability and Backup. The system is designed to handle errors gracefully without exposing sensitive information, corrupting data, or requiring a full system restart in most failure scenarios. DOST-STII administrators are responsible for implementing regular database backup procedures, particularly before and during multi-day events. Backup and restoration procedures are documented in the system documentation delivered with the completed application. The Docker-based deployment architecture also allows the server to be restored to a known-good state by restarting the containers from the last valid backup.



SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Marievic V. Narquita
Information Services Section (ISS) Head of DOST-STII

Date: April 8, 2026

